Carlos Manuel
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Portuguese.
Born 1962 in Luanda, Angola
CNRS Senior Scientist

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Education

1980–86 Bachelor in Physics, University of Lisbon. Master in Physics, University of Lyon I.
1986–89 PhD: Polymers at Interfaces. ENS Lyon, University of Lyon I.
1994 Habilitation. University of Strasbourg.

Professional Career

1987–89 Teaching assistant at the Ecole Normale Supérieure de Lyon.
1989–95 CNRS Research Scientist at Institut Charles Sadron, Strasbourg.
1989–90 Post Doctoral position, Cavendish Laboratory, Cambridge.
1995–96 Sabbatical leave, University of California in Santa Barbara.
1996–98 C.N.R.S.-Rhodia-Princeton University Complex Fluid Laboratory visiting fellow Princeton University
1998–22 CNRS Research Scientist, Institut Charles Sadron, Uni. Strasbourg. Senior Scientist since 2002.
2022– CNRS Senior Scientist, Chemistry Laboratory, Ecole Normale Supérieure de Lyon.

Awards

1994 Bronze CNRS Medal; *1999* CNRS Equipes Jeunes ACI; *2004* Alsace Research Award.

Teaching Experience

1987–89 Condensed Matter and Quantum Mechanics. ENS-Lyon and ULP Strasbourg.
1993 Self-Assembled Systems, Curso de Post-Graduação. USP, Brazil.
1993–95 Polymer Physics. Ecole Supérieure de Plasturgie, Oyonnax.
1998 Physics of Membranes, Short Course. Princeton University.
2004–19 A Random Walk in Soft Land. UNAM, Mexico. USP, Brazil. University of Strasbourg.
2000– Advanced Soft Matter, Kinetics, Self-Assembly, University of Strasbourg and ENS de Lyon.

PhD Supervision

23 PhD students supervised/cosupervised. 4 PhD students under supervision/cosupervision.

150^+ publications, choice of 8

[062] Impact of Polymer Tether Length on Multiple Ligand-Receptor Bond Formation. Jeppesen, C. *et al.*, *Science*, 2001, **293**, 465. A quantitative study of the role of ligand spacers on bio-adhesion.
[081] Photo-induced Destruction of Giant Vesicles in Methylene Blue Solutions. Caetano, W. *et al.*, *Langmuir*, 2007, **23**, 1307. First visualisation of photo-induced oxidation of lipid membranes.
[107] Gel-Assisted Formation of Giant Unilamellar Vesicles. Weinberger, A. *et al.*, *Biophys. J.*, 2013, **105**, 154. The easiest, fastest and most universal giant unilamellar vesicle's growing method.
[110] Enhanced Chemical Synthesis at Soft Interfaces: A Universal Reaction-Adsorption Mechanism in Micro-compartments. Fallah-Araghi, A. *et al.*, *P. R. L.*, 2014, **112**, 028301. Was life born in a droplet?
[113] Polymer collapse in miscible good solvents is a generic phenomenon driven by preferential adsorption. Mukherji, D. *et al.*, *Nature Communications*, 2014, **5**, 4882. A counterintuitive simple explanation.
[141] **The Giant Vesicle Book**, Ed. R. Dimova and C. Marques, Boca Raton, *CRC Press*, 2019.
[148] Accumulation of styrene oligomers alters lipid membrane phase order and miscibility. Morandi, M.I. *et al.*, *PNAS*, 2021, **118**, e2016037118. The hidden threat of plastic pollution.
[155] Activation energy for pore opening in lipid membranes under an electric field. Lafarge, E.J. *et al.*, *PNAS*, 2023, **120**, e2213112120. Biomembrane pores open under an electric field. Why?

Present Research

The science of lipid membranes: giant unilamellar vesicles as lipid bilayer platforms; lipid oxidation; pore formation; DNA, peptide and nanoparticle interactions with lipid membranes; bacterial oxidation.